SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING

COURSE CODE: CSA43

COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. (L2)

Assessment Tool: Code along exercise: Make your first HTML page together with CSS

Weightage: 10%

QUESTIONS: Total Marks: 10

Scenario: Web Development Standards Implementation

A start up is developing a modern web application. During development, the team encounters multiple issues related to HTML elements, HTTP methods, XML syntax, and web communication protocols.

The following observations are made:

- The team needs to embed a promotional video on the homepage.
- A registration form must securely send user data to the server.
- The navigation menu must follow semantic HTML standards.
- XML configuration files are being used but contain syntax errors.
- The application must communicate over the correct web protocol.

Data Snippets Provided

Snippet 1: Video Embedding Options

A. <media src="video.mp4"></media>
B. <video src="video.mp4" controls></video>
C. <movie src="video.mp4"></movie>
D. <vid src="video.mp4"></vid>

Snippet 2: Form Submission Methods

A. GET B. POST C. PUT D. DELETE

Snippet 3: Navigation Elements

A.   B. <nav> C. <section> D. <header>

Snippet 4: XML Syntax

A. <note><to>User</to><from>Admin</from>
B. <note><to>User</to><from>Admin</from></note>
C. <note><to>User</from></note>
D. <note to="User"></note

Snippet 5: Web Communication Protocols

A. FTP B. HTTP C. SMTP D. SNMP

1. Select the correct HTML tag for video embedding and justify your answer.
2. Identify the appropriate HTTP method for form submission and explain why.
3. Choose the proper semantic element for navigation and justify its usage.
4. Identify the correct XML syntax and explain the error in other options.
5. Choose the correct protocol for web communication and justify your choice.

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

ANSWERS

Answer 1: Video Embedding (Snippet 1)

Correct answer: B. <video src="video.mp4" controls></video>

Justification. <video> is the only standard element that HTML5 provides for embedding video. It plays the file natively in the browser without plugins such as Flash. The *src* attribute gives the file location and the Boolean *controls* attribute shows the browser's play, pause, volume and full-screen controls. Further attributes such as *poster*, *preload*, *autoplay*, *muted* and *loop* can be added, and nested <source> and <track> elements provide alternative formats and captions.

Why the others are wrong. <media> (A), <movie> (C) and <vid> (D) are not defined in the HTML specification. Browsers treat them as unknown elements and display nothing, so the promotional video would not appear.

Answer 2: Form Submission Method (Snippet 2)

Correct answer: B. POST

Justification. A registration form sends personal data such as name, email and password. POST places this data in the request body, so it does not appear in the URL, browser history, bookmarks or server logs. POST has no practical size limit and is meant for requests that create or change data on the server, which is what registration does. Combined with HTTPS, it protects the data in transit. POST is not encrypted by itself, so HTTPS is still required.

Why the others are wrong. GET (A) appends data to the URL, exposing it and limiting its length, and it is intended only for reading data. PUT (C) replaces a resource at a known URL and DELETE (D) removes one, and HTML forms do not support either method natively (only GET and POST), though they can be used through JavaScript or APIs.

Answer 3: Semantic Navigation Element (Snippet 3)

Correct answer: B. <nav>

Justification. <nav> is the semantic HTML5 element for a block of major navigation links, such as the main menu. Its meaning is understood by browsers, search engines and assistive technology, so screen reader users can jump directly to the navigation landmark. It makes the code readable and improves SEO and accessibility.

```
<nav>
  <ul>
    <li><a href="/">Home</a></li>
    <li><a href="/register">Register</a></li>
    <li><a href="/contact">Contact</a></li>
  </ul>
</nav>
```

Why the others are wrong. (A) is a generic container with no meaning. <section> (C) groups thematic content and is not specifically for navigation. <header> (D) holds introductory content or branding and may contain a nav, but it is not itself the navigation element.

Answer 4: Correct XML Syntax (Snippet 4)

Correct answer: B. <note><to>User</to><from>Admin</from></note>

Justification. Option B is well formed. Every start tag has a matching end tag, the elements are properly nested (to and from are closed inside note), and there is a single root element, note.

Errors in the other options.

Option	Error
A	The root element <note> is never closed. The closing </note> tag is missing, so the document is not well formed.
C	Tags are mismatched: <to> is closed with </from>. XML requires an end tag with the same name, and elements must be nested correctly.
D	The closing tag is malformed: </note is missing the final >. (The attribute value is correctly quoted, but the tag is not complete.)

XML is case-sensitive and strict. Unlike HTML, a single syntax error makes parsers stop with a fatal error, so configuration files must be well formed.

Answer 5: Web Communication Protocol (Snippet 5)

Correct answer: B. HTTP

Justification. HTTP (HyperText Transfer Protocol) is the application-layer, request-response protocol used by browsers and web servers to transfer web pages, images, scripts and API data. It runs over TCP (or QUIC for HTTP/3) and, secured with TLS as **HTTPS**, it also gives encryption and server authentication. The application in the scenario should use HTTPS in production.

Why the others are wrong. FTP (A) transfers files between hosts and is not used for delivering web pages to browsers. SMTP (C) sends email between mail servers. SNMP (D)

is used for monitoring and managing network devices.

Summary of Answers

No.	Topic	Answer	Key Reason
1	Video embedding	B. <video>	Standard HTML5 element with native playback controls
2	Form submission	B. POST	Data in body, not exposed in the URL
3	Navigation	B. <nav>	Semantic landmark for major navigation links
4	XML syntax	B	Well formed: matched tags, proper nesting, one root
5	Web protocol	B. HTTP	Application protocol for web content delivery

Code-Along: First HTML Page with CSS

The following page applies all five correct choices together. It uses a semantic structure, a <video> element, a POST form and an external stylesheet.

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>StartUp Home</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <header>
    <h1>StartUp</h1>
    <nav>
      <a href="/">Home</a>
      <a href="#register">Register</a>
    </nav>
  </header>
  <main>
    <section>
      <h2>Our Promo</h2>
      <video src="video.mp4" controls width="480"></video>
    </section>
    <section id="register">
      <h2>Register</h2>
      <form action="/register" method="POST">
        <label for="email">Email</label>
        <input type="email" id="email" name="email" required>
        <button type="submit">Sign Up</button>
      </form>
    </section>
  </main>
  <footer><p>&copy; 2026 StartUp</p></footer>
</body>
</html>
```

style.css

```
body   { font-family: "Times New Roman", serif; margin: 0; }
header { background: #1f3a5f; color: #fff; padding: 12px 24px; }
nav a  { color: #fff; margin-right: 16px; text-decoration: none; }
main   { padding: 24px; }
section { margin-bottom: 24px; }
video  { max-width: 100%; }
input, button { padding: 6px 10px; margin: 4px 0; }
footer { text-align: center; padding: 12px; background: #eee; }
```